Lucas Zheng

408-438-0722 | lucas_zheng@brown.edu
linkedin.com/in/lucas-j-zheng | github.com/lucas-j-zheng | lucas-j-zheng.github.io/

Education

Brown University Providence, RI
B.S. in Computer Science & B.A. in Applied Mathematics; GPA: 4.00/4.00 *Expected May 2027*
Sc.M. in Computer Science *Expected May 2028*

- Coursework: Statistical Inference, Machine Learning, Deep Learning, Sequential Decision Making, Algorithms

Experience

Incoming Quantitative Trading Intern Jun. 2027 – Aug. 2027
Citadel Securities *New York, NY*

Research Assistant Sep. 2025 – Present
Intelligent Robot Lab, Brown University *Providence, RI*

- Developed an option-level R-MAX planner with prioritized sweeping over 28 temporally extended skills in Montezuma's Revenge (a sparse-reward, long-horizon Atari benchmark), reaching 22 of 24 rooms and scoring 6,800
- Designed a compact RAM-based state abstraction for tabular planning, discovered that masking laser timers caused non-Markov states, and restored reliable behavior by selectively retaining laser-timer state in affected rooms
- Built an option-level DQN with per-option Coin Flip Network heads to estimate state-option pseudocount bonuses, masking unavailable options at each state

Applied Machine Learning Intern Jun. 2026 – Aug. 2026
Shopify *New York, NY*

- Built an end-to-end, locale-aware evaluation system to gate AI-generated sales-outreach emails, spanning labeled eval-set generation, LLM-judge scoring, and a feedback loop turning reviewer annotations into prompt revisions
- Extended the gate to 6 non-U.S. locales covering hundreds of thousands of outreach emails per year, improving judge F1 against human-labeled ground truth from 0.57 to 0.74 through model and context-rotation upgrades

Software Engineering Intern Jul. 2025 – Aug. 2025
Supio *Seattle, WA*

- Reduced deposition transcript turnaround time from 10 days to minutes by engineering an end-to-end pipeline with Whisper, pyannote speaker diarization, and an OpenAI API correction workflow
- Integrated transcripts into a hybrid RAG system combining pgvector semantic search with BM25 lexical ranking, enabling lawyers to query hours of audio with sub-second response times

Projects

Steering Diffusion Skills for Reliable Skill Chaining Sep. 2026 – Present

- Developing a framework for a symbolic (PDDL) planner to chain frozen diffusion-policy skills in CALVIN, steering each skill's sampled actions toward subgoal states from which the next skill succeeds
- Training diffusion policies to produce more diverse successful trajectories without lowering success rate, giving the steerer more compatible handoff states for downstream skills

2026 U.S. Senate Election Forecast Jul. 2026 – Present

- Forecast all 35 races by blending a ridge fundamentals prior with a weighted poll average treated as a pseudo-poll, then running 40,000 Monte Carlo simulations with correlated, fat-tailed national, regional, and state errors
- Backtested on held-out 2018–2024 election cycles, achieving a 0.071 Brier score versus 0.082 for the fundamentals prior and 0.085 for polls alone

Technical Skills

Programming: Python, TypeScript, SQL
Tools & Frameworks: PyTorch, NumPy, pandas, scikit-learn, Matplotlib, Jupyter, Gymnasium, PostgreSQL, Git

Miscellaneous

Hackathons: 2nd Place, 2026 Precision Neuroscience BCI Hackathon
Spoken Languages: Mandarin (fluent), Arabic (working proficiency – U.S. State Department NSLI-Y Scholar)
Interests: Squash, Golf, Poker, Chess